

Carys J. E. Gilbert

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EDUCATION

PhD - Astrophysics | 2026-

University of Manchester, United Kingdom

- Thesis Title: Tracing the Epoch of Reionisation using Galaxies and Intergalactic Gas with Euclid and the SKA
- Primary Supervisor: Prof Chris Conselice (UoM)

MSc. by Dissertation - Astronomy | 2025-2026

University of Cape Town, South Africa

- Thesis Title: X-Ray and Full-Spectrum SED Analysis of 12MGS Galaxies in the Southern Sky
- Supervisors: Dr Lucia Marchetti (UCT), Dr Luigi Barchiesi (UCT), Prof. Carlotta Grupponi (INAF-OAS), and Prof. Mattia Vaccari (UCT)

BSc. Hons. - National Astrophysics and Space Science Programme | 2024

University of Cape Town, South Africa

- Graduated with distinction (87%) for both the coursework and dissertation components
- Awarded the AST4007W Class Medal for the highest mark of all students in the 2024 NASSP Programme

BSc. - Astrophysics and Physics | 2021-2023

University of Cape Town, South Africa

- Graduated with distinction (79%) in Astrophysics

PUBLICATIONS

***A New Hope* for AGN SED Fitting: X-Ray Spectral Analysis of 12MGS AGN with the C2PO-Torus Model**

C.J.E. Gilbert et al. (2026)

Accepted by Monthly Notices of the Royal Astronomical Society (MNRAS)

arXiv: <https://arxiv.org/abs/2609.38350>

Beyond the Dot: An LRD-like Nucleus at the Heart of an IR-bright Galaxy and its Implications for High-redshift LRDs

P. Rinaldi, G.H. Rieke, Z. Wu, **C.J.E. Gilbert** et al. (2026)

The Astrophysical Journal, Volume 1006, Number 2

DOI: [10.3847/1538-4357/ae80cd](https://doi.org/10.3847/1538-4357/ae80cd)

Filling the Gap in Cluster Evolution: JWST's Glimpse into a Young, Star-Forming Cluster at Cosmic Noon

P. Rinaldi, S. Alberts, C.N.A. Willmer, C. Carreira, C.C. Williams, G. Noirot, **C.J.E. Gilbert** et al.

Submitted to ApJ, under review

arXiv: <https://arxiv.org/abs/2602.24162v1>

SUCCESSFUL PROPOSALS

Nuclear Spectroscopic Telescope Array (NuSTAR) Cycle 11 | 2025

IRAS 03450+0055: An Unexplored Local Mid-Infrared Bright AGN

- Proposal Number 11077, **P.I. C. Gilbert**, 20ks awarded
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RESEARCH PROJECTS

X-Ray and Full-Spectrum SED Analysis of 12MGS Galaxies in the Southern Sky

Master's Thesis, 2025-2026

- Continuing the work of my Honours thesis by repeating the analysis for the remaining 31 12MGS AGN within the Southern Sky for which we have full MeerKAT and SALT coverage
- Completing X-ray spectral analysis with physically motivated models for the entire sample of 43 sources to determine intrinsic AGN parameters
 - Visited Prof. Cristian Vignali at the University of Bologna in June 2025 to learn about physical modelling techniques
- Completed radio spectral analysis using the MeerKAT data to complete multiwavelength coverage
- Fitted full-spectrum SEDs using the CIGALE software

X-Ray and SED Analysis of Local 12 μ m Galaxies

Honours' Research Thesis, 2024

- Supervisors: Dr Lucia Marchetti and Dr Luigi Barchiesi
- Used archival Chandra and NuSTAR data to produce X-ray spectra of 12 galaxies selected from the 12MGS galaxy sample
- Determined the intrinsic X-ray luminosity of the central AGN using XSPEC
- Used the results to produce SED fits of the galaxies using the CIGALE software
- Found that including the X-ray data increased the quality of the fits, especially for obscured AGN, and allowed greater precision in determining galactic parameters such as stellar mass and AGN fraction
- Achieved 96% overall

Long-Term Study of the Gamma-Ray Binary LS I +61° 303

3rd-Year Physics Capstone Project, 2023

- Supervisor: Dr Itumeleng Monageng
- Re-reduced and analysed optical spectra spanning 3 years from the Liverpool Telescope to find radial velocities of the system
- Used Keplerian fitting to determine the system's orbital parameters. Ultimately concluded that the companion is likely to be a black hole given the unknown orbital inclination
- Achieved 78% overall

CONFERENCES AND PRESENTATIONS

SARAO Postgraduate and Postdoctoral Research Conference | 2025

- Presented a poster on my Master's research

ISARP RADIOMAP Workshop | 2025

- Member of the LOC - <https://idia.ac.za/isarp-radiomap-2025-soc-loc/>

University of Bologna - Extragalactic Research Group | 2025

- Presented ongoing research at the weekly extragalactic research group meeting at the University of Bologna

AWARDS

- 2026 University of Manchester Faculty of Science and Engineering African Futures Doctoral Scholarship
 - 2025 SA College Croll Scholarship awarded for achieving an Honours degree with distinction
 - South African Radio Astronomy Observatory 2024 Honours Scholarship and 2025 Master's Scholarship
 - 2024 AST4007W Class Medal for the highest overall mark
 - Astronomical Society of Southern Africa's 2023 Cooke Scholarship for Studies in Astronomy
 - Dean's Merit List for achieving a weighted course average above 70%, University of Cape Town, 2022-2023
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WORK EXPERIENCE

Course Tutor for NASSP Honours | 2025

University of Cape Town

- Tutor for the Observational Techniques 2 - Radio Astronomy module for the National Astrophysics and Space Science Program Honours course
- Responsible for organising and running weekly tutorial sessions
- In charge of marking all assignments

Course Tutor for 2nd Year Astrophysics | 2025

University of Cape Town

- Tutor for the AST2004H 2nd-Year Astrophysics course
- Responsible for organising and running weekly tutorial sessions
- In charge of marking all assignments and the oral examinations

Internship at SAAO/SALT | 2023

Completed a 3-week internship at the South African Astronomical Observatory

- Main project was the conversion of the Java Visibility tool to a Python library
- Culminated in a 3-day excursion to Sutherland and the Southern African Large Telescope to observe operations and changing of instruments

Private Tutoring | 2021 Onwards

- Tutoring High School Mathematics, Physical Sciences and Accounting

TEACHING AND SUPERVISING EXPERIENCE

Co-Supervisor for Honours Project | 2025

- Co-supervised an Honours student, Christiaan Schutte, during his 4th-year research thesis

Guest Lecturer for 2nd-Year Astrophysics | 2025

- Taught the "Milky Way" module for the 2nd-year AST2003H astrophysics course

VOLUNTEERING AND OUTREACH

Volunteer Coordinator for Soapbox Science Cape Town | 2024-2026

Part of the volunteering team for the 2024 event and volunteer coordinator in 2025

- In charge of selecting, training and coordinating all volunteers for the 2025 event
- Conducted visitor counts and surveys throughout the event, and assisted with setting up and closing down the event, attracting visitors, and with speaker presentations and demonstrations

Curiosity Cube Demonstrator | 2025

Merck Curiosity Cube Demonstrator during UCT Science Outreach Week 2025

- Ran the demonstrations for groups of ~30 Grade 7 learners at a time
- Assisted with the setting up and closing down of the Cube each day

Ambassador and Volunteer at the International Astronomical Union General Assembly | 2024

Part of the IAU GA 2024 volunteer team based in Cape Town

- Assisted with public relations, technical equipment, and management of delegates
- Responsible for the allocation and organisation of e-posters across the entire conference
- Was in charge of the conference Virtual Reality booth

Speaker for Women in Tech South Africa | 2024

- Was invited to give a presentation on South African astronomy and my journey as a woman in STEM, alongside the SpaceX astronaut Dr Sian Proctor

Presenter for Girls in Robotics Africa | 2024

- Was invited to give a presentation on astronomy and spectroscopy to girls aged 12-17 as part of their STEM education and mentoring program
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LEADERSHIP POSITIONS AND EXPERIENCE

Co-Chair of the UCT Astronomy Outreach Society | 2025

- Help to organise and run the open days and job shadowing programs for local high school students

Vice-President of the UCT Society for Physics Students | 2023

- Organised social and educational events for students and members of staff

Class Representative | 2023

- PHY3004W: Advanced Physics
- AST3002F: Stellar Astrophysics
- AST3003S: Galactic and Extragalactic Astronomy

SKILLS AND ABILITIES

- Proficient in Python, including the Astropy, scipy and matplotlib libraries
- Experience in conducting observations using optical, spectroscopic, and radio telescopes
- Experience in processing and analysing optical (photometric and spectroscopic), radio and X-ray astronomical data
- Completed the National Institute for Theoretical and Computational Science (NITheCS) MUST 5 Day Deep Learning Bootcamp
- Proficient in LaTeX

REFERENCES

University of Cape Town

- Dr Lucia Marchetti (Senior Lecturer) - lucia.marchetti@uct.ac.za
- Dr Luigi Barchiesi (Post-Doc) - luigi.barchiesi@uct.ac.za
- Dr Itumeleng Monageng (Senior Lecturer) - itumeleng.monageng@uct.ac.za

University of Bologna

- Prof Cristian Vignali (Associate Professor) - cristian.vignali@unibo.it

SAAO

- Prof Rosalind Skelton (SAAO Managing Director) - ros@sao.co.za
 - Dr Lisa Crause (SALT Observatory Scientist) - lisa@sao.co.za
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